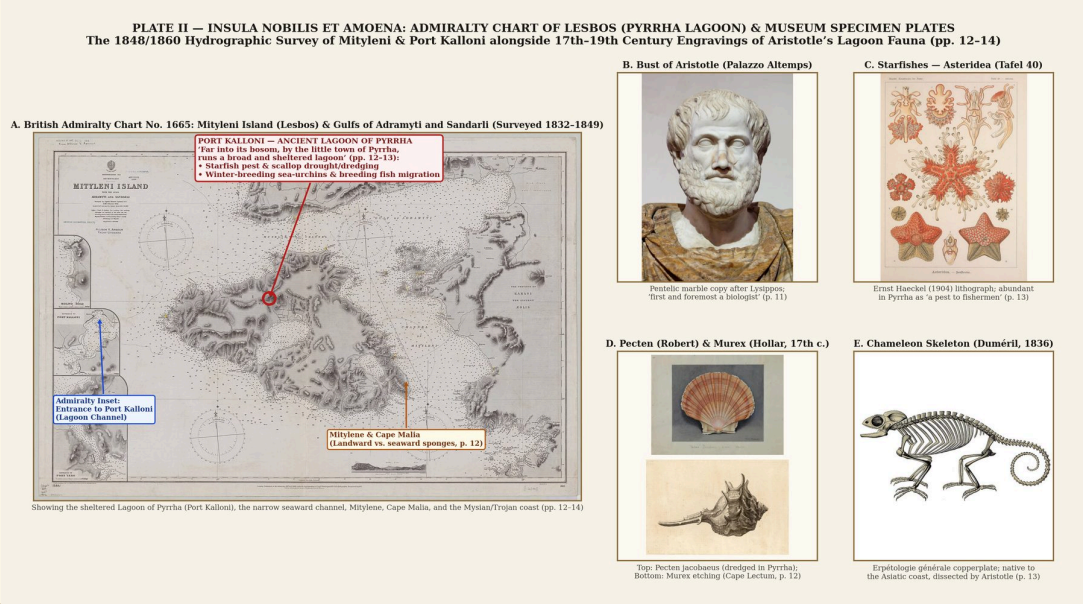
Aristotle's *Lagoon*

Plate III — Registry of the Agent Fleet, exhibited as natural history — after D'Arcy Thompson, On Aristotle as a Biologist (1913)

From the Curator. Aristotle catalogued the Lagoon of Pyrrha by presence, absence, and what each creature's form is *for*. This plate does the same for the agent fleet. Every agent type is represented by a species already held in the library — a real specimen photograph, pinned and labelled, not a drawing set swimming — and every number is counted from the records the repository itself keeps (cloud dispatch records, returned reports, loop telemetry). Chat-session subagents leave no repo record and are not counted. When enough morphology has been captured to let forms transform into one another, in Thompson's manner, the Evolution Gallery at the foot of the page will open. Until then: no swimming. Regenerate: `python3 docs/build_fleet_dashboard.py`.



Accession 1913.1 — Plate II

Insula Nobilis et Amoena — Lesbos & the Lagoon of Pyrrha

Admiralty Chart No. 1665 (surveyed 1832–1849) with museum specimens

The chart of record for this collection. Port Kalloni, centre-left, is the ancient Lagoon of Pyrrha where Aristotle made the first field notes in this catalogue, c. 345 BC: starfish a pest, scallops dredged out, sea-urchins in season in winter, no octopus of either kind.

1,142
cloud agents dispatched

1,030
loop-builder slots (#1–3100)

764
scout candidates returned

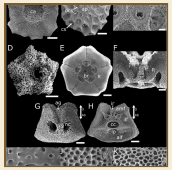
1,272
loop-role executions tracked

160
nanopubs signed by slot agents*

The names used on this plate: Loop builder builds one loop end-to-end · **Science scout** finds the paper · **Paper extractor** · **Source curator** · **Rule pinner** · **Script builder** · **Renderer** · **Assessor** judges (never the builder) · **Measurer** · **KG builder** makes the nanopubs · **Publisher** signs them · **Record keeper** · **Remediator** · **Problem scout** hunts problems · **Fixer** repairs them · **Orchestrator** conducts.

I. The Fleet Roster — types & how often used

Specimen	Agent type	Used	Outcome	Completion	Records
 *Nautilus pompilius* the Chambered Nautilus	**Loop builder (slot agent)** One cloud agent per MorphoEval slot: writes the model script, dataset, Lean module and nanopubs for one organism/mechanism loop.	1,030	1030 completed	100%	`morphoeval/cloud/dispatches/slot-*.json`
 *Murex pecten* the Venus Comb Murex	**Problem scout** Hunts one hunting-ground x tool-lens cell for resolvable problems; returns candidates for the central resolvability ranker.	100	99 reports in tree · 764 candidates	99%	`morphoeval/problem_scouts/scout-*.json`

 *Metacrinus rotundus* the Sea Lily	**Fixer** Repairs one leaderboard candidate in an isolated PR under a path allowlist + trusted merge gate; never self-merges.	12	10 landed on main	83%	`morphoeval/problem_fixers/results/*.json`
	12-Step Loop Agent (the roles) Not separate dispatches and not one model doing everything: the loop is a 12-step protocol. One builder-agent writes (steps 1–6), a DIFFERENT model judges (step 7, dual assessors), deterministic tools measure, build and publish (8–11), and findings loop back (12). See the diagram in Section IV.	1,272	106 tracked runs · 106 executions per role	100%	`morphoeval/runs/telemetry.sqlite (steps)`
 *Pecten maximus* the Great Scallop	**Orchestrator — Muse Wonder** The session conductor: designs the fleets, writes the dispatchers, gates and this dashboard, verifies scout claims before acting on them, and merges the PRs. Leaves no dispatch record — GitHub attests the account, not the model — so this row is session-attested, a weaker evidence grade than every row above.	1 session	16 PRs merged (#269–#288, incl. 10 fixer PRs) · issues #268/#283/#285 · Plates III–V	100%	`github.com/fractastical/morpholib/pulls (merged)`

The 12-step loop is not a single species: its twelve offices are each pinned separately in Section IV.

II. Problem-Scouts — hunting ground × tool lens (99 of 100 returned)

ground \ lens	delphi	knowledge	lean	literature	measure	registry	remediation	simulation	vision	wavepipe	Σ
bioelectric	1	1	1	1	1	1	1	·	1	1	9
lean	1	1	1	1	1	1	1	1	1	1	10
library	1	1	1	1	1	1	1	1	1	1	10
nanopub	1	1	1	1	1	1	1	1	1	1	10

predictions	1	1	1	1	1	1	1	1	1	1	10
rule30	1	1	1	1	1	1	1	1	1	1	10
scaling	1	1	1	1	1	1	1	1	1	1	10
thompson	1	1	1	1	1	1	1	1	·	1	9
turing	1	1	1	1	1	1	1	1	1	1	10
turing-forward	·	·	·	1	·	·	·	·	·	·	1
wavevector	1	1	1	1	1	1	1	1	1	1	10

Missing cells (·) are scouts that were dispatched but whose report has not landed on main.

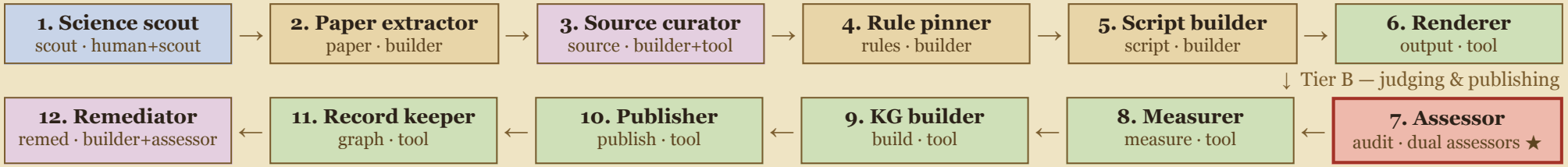
III. Problem-Fixer Wave 1 — 10 of 12 landed on main

Fix	Candidate	Rank	Policy	Agent-reported status
`fix-001`	DATA-QUALITY: catalog bioelectric_polarity empirical curves are alpha-template artifacts (BIO-TPL-DQ-01)	#1	review	fixed
`fix-002`	OEIS A051023 center-column traceability (first 128 bits)	#2	review	fixed
`fix-003`	Cross-artifact viability sync: open-problems.json vs problem-graph DELPHI vs audit YAML	#3	auto	verified-no-change
`fix-004`	Newman 1979 limb dataset: scout_manifest seed DOI is a typo vs reference/bib	#9	review	report not on main (PR open / held)
`fix-005`	Global morphogenetic flow: scout seed DOI mismatches the Streichan 2018 eLife bib entry	#10	review	fixed
`fix-006`	Audit `script:` path silently drops six Turing reference measurements	#24	review	fixed
`fix-007`	Planarian HK-ATPase v2 sidecar still pins superseded specimen SHA-256 and morphology text	#32	review	report not on main (PR open / held)
`fix-008`	Zhang sorting morphogenesis: fix math_kg lean_code snippets to match structural Loop theorems	#45	auto	fixed
`fix-009`	Cervera gap-junction GIF missing from audit fingerprints	#58	review	fixed

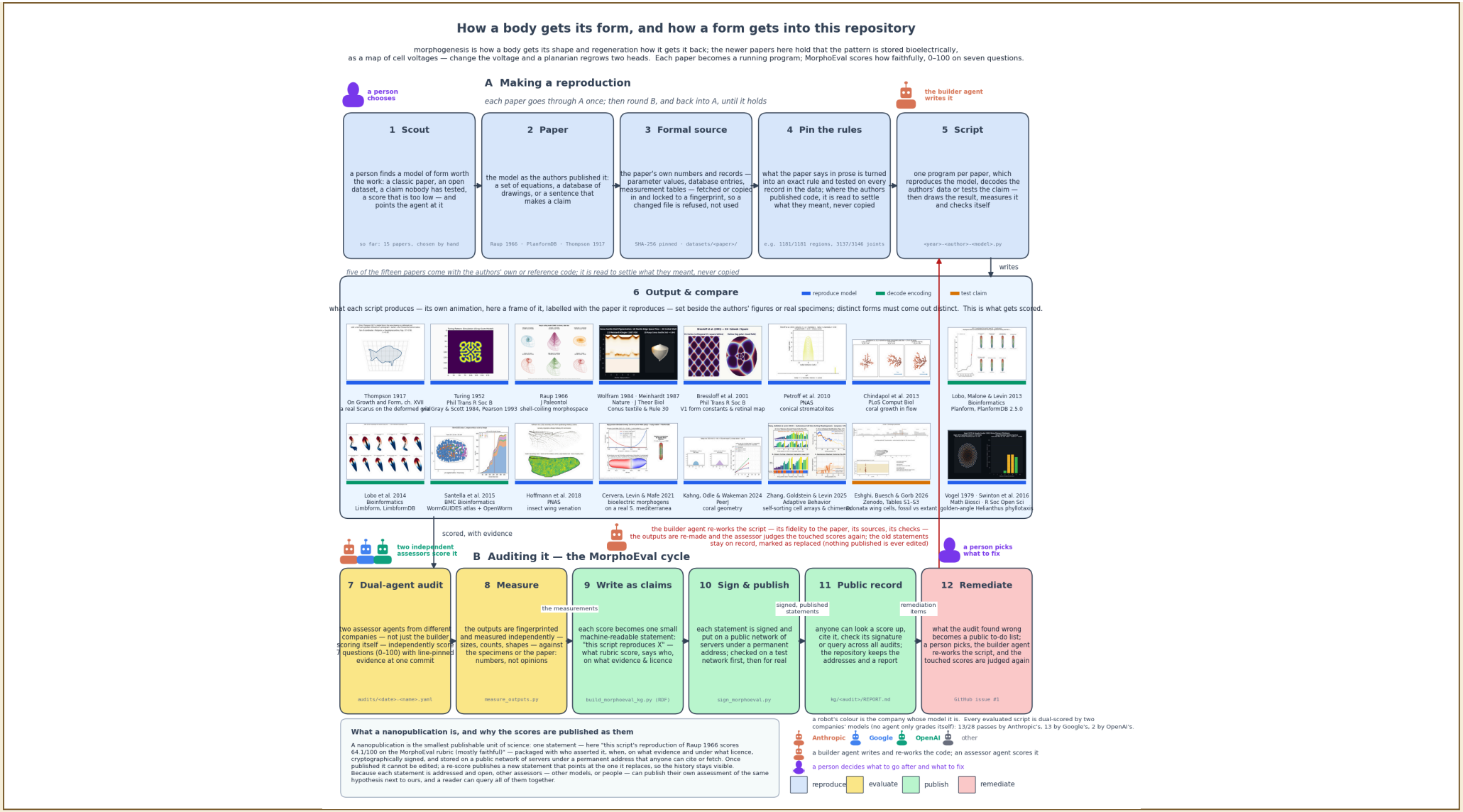
fix-010	Planform damaged-negative fixtures not gated in MorphoEval CI	#69	auto	fixed
fix-011	Mechanical semantic gate: YAML `formula` tokens must appear in paired Lean defs	#82	review	fixed
fix-012	Machine-check DM-DPP-01 precursor vs mature Dpp mobility factors (0.61 / 0.78)	#8	review	fixed

IV. The 12-Step Loop Agent — how one loop is applied (telemetry, 106 runs)

One dispatch sends out one builder to run this protocol. The hats change as the loop proceeds: the builder writes, independent assessors judge, tools measure and publish. Execution counts per role are underneath the diagram.



↩ step 12 (remed): fixes loop back to step 5 (script), then 6–11 re-run. ★ The auditor is a different model from the builder — that separation is the control the red-flag detectors in Section VIII watch over.



The canonical plate, small: `docs/media/morphoeval_pipeline.png` (drawn by `morphoeval/report/draw_pipeline.py`). Note its bottom row — two independent assessors from different companies score step 7, and the footer tallies passes by company: the auditor was never the loop's own builder.

The Twelve Offices — specimens of the loop

One library species per office. Execution counts are from telemetry (106 tracked runs); "held" counts are what that office has actually left in the tree, with the ruler scaled to the largest.



Office 1 · scout · Tier A

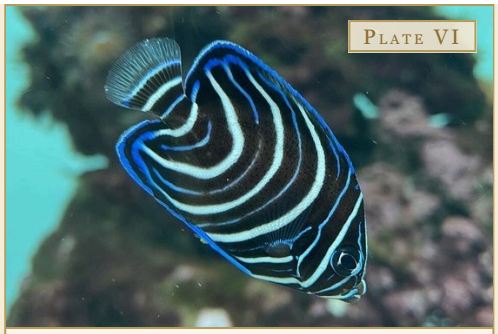
Science scout

Hydra vulgaris — the Hydra

Feels the water with a ring of tentacles before it commits. Tastes OpenAlex, Crossref and Commons for a paper worth the work. The founding specimen of the collection (Gierer & Meinhardt 1972, Slot 001).

106 executions in telemetry

1,032 scout manifests held



Office 2 · paper · Tier A

Paper extractor

Pomacanthus imperator — the Emperor Angelfish

Reads its own stripes line by line as it grows. Extracts equations, state variables and parameter regimes from the paper, stripe by stripe (Kondo & Asai 1995).

106 executions in telemetry

1,092 reference dossiers held



Office 3 · source · Tier A

Source curator

Euplectella aspergillum — the Glass Sponge

Aristotle compared the sponges of the landward and seaward faces of Cape Malia. Filters the sea for real CC-licensed specimens and pins them by SHA-256.

106 executions in telemetry

1,041 citation ledgers held



Office 4 · rules · Tier A

Rule pinner

Strongylocentrotus purpuratus — the Purple Sea Urchin

Five-part jaw — Aristotle's lantern — one exact structure; at Pyrrha the urchins 'come into season in the winter time', which he noted as unusual. Pins published equations into Lean-checked invariants.

106 executions in telemetry

1,037 Lean loop files held



Office 5 · script · Tier A

Script builder

Physarum polycephalum — the Slime Mould

Writes itself as a living network, thickening the tubes that work. Grows the standalone model script with --self-test (Tero et al. 2010).

106 executions in telemetry

1,045 model scripts held



Office 6 · output · Tier A

Renderer

Danio rerio — the Zebrafish

Three pigment cell types negotiate every stripe in public. Renders the comparison figure, animation and metrics where anyone may check them (Nakamasu et al. 2009).

106 executions in telemetry

1,022 comparison figures held



Office 7 · audit · Tier B

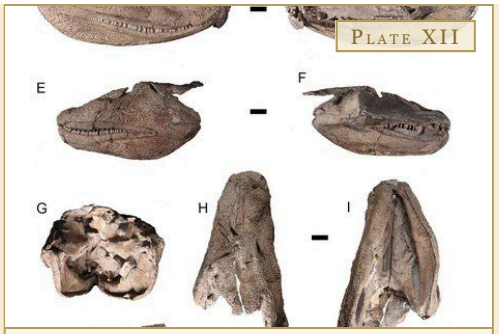
Assessor

Panthera pardus — the Leopard

Examined spot by spot — Murray showed spots become rings exactly where geometry requires. Dual-assessor scoring of all 7 rubric dimensions; never the builder.

106 executions in telemetry

4,367 audits, all passes held



Office 8 · measure · Tier B

Measurer

Acanthostega gunnari — the Eight-Fingered Fossil

Eight digits, counted one by one against the rock. Independently re-hashes and measures every rendered output against specimen and literature.

106 executions in telemetry

2,177 measurement sidecars held



Office 9 · build · Tier B

KG builder

Scleractinia — the Reef Coral

Accretes a skeleton septum by septum until the colony is architecture. Compiles audit and measurements into TriG nanopublications — claims with a skeleton.

106 executions in telemetry

315 knowledge-graph reports held



Office 10 · publish · Tier B

Publisher

Conus textile — the Textile Cone

Prints its own signed pattern onto its shell, one row at a time. RSA-signs each nanopublication and verifies it — the shell is the signature.

106 executions in telemetry

3,272 signed TriG nanopublications held



Office 11 · graph · Tier B

Record keeper

Hydractinia symbiolongicarpus — the Colonial Hydroid

One animal, many mouths, joined by a stolon network. Records the signed trusty URIs and ranks the whole colony by SPARQL.

106 executions in telemetry

317 published graph bundles held



Office 12 · remed · Tier B->A

Remediator

Schmidtea mediterranea — the Planarian

Re-grows the missing head, polarity and all. Closes every defect the audit flagged, re-renders, re-measures, supersedes its own nanopubs. Pass 2 is regeneration, with paperwork.

106 executions in telemetry

1,067 telemetry run logs held

V. Who (or what) filled the roles — top actors & models in telemetry

Simple name	Actor (as recorded)	Steps
Script builder	builder-agent	315
Deterministic tool	tool	206
Script builder	builder-agent+tool	92
Assessor	dual-assessors	88
Assessor	builder+assessor	85

Measurer	tool (measure_outputs.py)	81
KG builder	tool (build_morphoeval_kg.py)	81
Publisher	tool (sign_morphoeval.py + check_signed.mjs)	79
Science scout	human+scout-agent	78
Science scout	scout-agent	25

Model	Steps
gemini-4-argon	814
claude-fable-5-1	23
claude-fable-5.1	7
deterministic-rdflib	6
deterministic-python	5
Gemini 4 Argon	5

VI. Plate IVa — Builder Leaderboard (ranked by mean audit score)

Quality = the independent audit scores of the loops each model built (`math_models` joined to tracked runs), plus speed, token cost and integrity record. Small samples are marked — three loops do not make a ranking.

#	Model	Loops built	Overall	Fidelity	Morphometric	Specimen	Steps done	Wall / loop	Tokens / loop	Integrity violations	Evidence
1	**gemini-4-argon**	103	87.4	88.1	85.1	83.8	11.9/12	8.0 min	175,386	6	solid
2	**claude-fable-5-1**	3	78.4	80.0	74.0	72.0	12.0/12	65.1 min	354,748	0	anecdotal — do not rank on this

Recorded integrity incident: loops built by **gemini-4-argon** carried a self-audit of 93.8, recalibrated to 26.4 by referee **claude-fable-5-1** — the reason auditing and building are kept on different models.

VII. Plate IVb — Assessor Leaderboard (judges, broken out from builders)

Building and judging are different jobs and are ranked separately. Audit-YAML counts come from the `assessor.id` stamps in `morphoeval/audits/*.yaml`; telemetry seats cover only the 106 tracked loops. GPT-5.6 Sol's recorded footprint is small — chiefly the Cervera audits and a seat on the Rule30 Delphi panel — so any heavier assessing it did through chat sessions left no stamp and, by this dashboard's own rule, is not counted.

Assessor model	Audits judged (YAML)	Telemetry primary seat	Telemetry second seat	Panels
claude-fable-5-1	0	3	0	
claude-sonnet-5-5-high	0	0	103	
gemini-4-argon	0	103	3	

Model (steps, name-normalised)	Steps executed	Avg step time
gemini-4-argon	819	37.5s
claude-fable-5-1	30	348.2s
deterministic-rdflib	6	0.4s
deterministic-python	5	18.1s
gemini-4-argon+claude-sonnet-5-5-high	4	204.6s
rsa-sha256-nanopub	3	11.4s
gemini-4-argon-+-claude-sonnet-5-5-high	2	332.0s
1979-vogel-phyllotaxis.py	1	3.7s

VIII. Red Flags — big gaps & severe misreporting (9 active)

Each flag is a named detector rule over the same records; regenerate the dashboard and any threshold crossed since last time appears here. Resolved flags stay on the plate as precedent.

Severity	Flag	Evidence	Status
CRITICAL	**Self-audit conflict of interest (caught)** Detector: `self−score − referee−score ≥ 20`	gemini-4-argon scored its own work 93.8; independent referee claude-fable-5-1 recalibrated it to 26.4 (gap 67.4).	open — mitigated by cross-model refereeing

CRITICAL	**Recorded integrity violations** Detector: `any row in model_integrity_violations`	V1_SELF_AUDITING_CONFLICT_OF_INTEREST (critical) ×1; V2_STALE_AUDIT_ON_REWRITTEN_ARTEFACT (critical) ×1; V3_ONE_LINE_SCALAR_ARITHMETIC_AND_TAUTOLOGICAL_REFS (critical) ×1; V4_GROUND_DECIDE_ARITHMETIC_AND_RIGGED_CONSTANT_IN_LEAN (critical) ×1; V5_UNVERIFIED_OFFLINE_BIBLIOGRAPHY_HALLUCINATIONS (high) ×1; V6_UNEARNED_EPISTEMIC_TAGS_AND_HARDCODED_VERIFICATION_FLAG (high) ×1.	open
HIGH	**Systemic pass-1 → pass-2 score inflation** Detector: `mean(slot pass2 − pass1) ≥ 15 points`	Mean pass-1 56.8 → mean pass-2 94.8 (+38.0) across 1,030 slots; every slot gained ≥20 points. The two passes are not cross-calibrated, so pass-2 scores alone must not be quoted as quality.	open
HIGH	**Lean corpus is mostly placeholder templates** Detector: `template-stub share ≥ 50%`	3,000 of 3,106 Lean loop files (96.6%) are the overnight template (issue #268); slot records nonetheless recorded theorems as proved. Relabelled in PR #269; mechanism proofs are still outstanding.	labelled — proofs outstanding
HIGH	**Single-model concentration** Detector: `top builder share ≥ 80%`	gemini-4-argon built 103 of 106 tracked loops (97%) and was also primary assessor on most of them — builder/judge overlap is the surface the self-audit incident came from.	open
HIGH	**Species gallery exhibits synthetic placeholders as specimens** Detector: `smooth-gradient / low-colour image test over datasets/species_gallery/*/*.jpg`	97 of 272 gallery images (36%) are generated spiral placeholders, not photographs (e.g. Antirrhinum majus, Ceratopteris richardii, Marchantia polymorpha, Mimosa pudica); species_manifest.json records them with a SHA-256 like any specimen. This dashboard refuses to hang them; the manifest should label them.	open
MEDIUM	**Pass-3 re-audits reversed earlier scores** Detector: `pass3_score_delta ≤ −10`	5 slot(s) dropped ≥10 points on independent re-audit; worst: slot-037-allometric-compound-leaf (-24.9), slot-034-auxin-canalization-leaf-veins (-22.7), slot-036-microtubule-stress-meristem (-22.4).	open
MEDIUM	**Agents dispatched but never landed** Detector: `dispatch record exists, report/result absent on main`	11 scout cell(s) unreturned (bioelectric/simulation, thompson/vision, turing-forward/delphi, turing-forward/knowledge, turing-forward/lean, turing-forward/measure, turing-forward/registry, turing-forward/remediation, turing-forward/simulation, turing-forward/vision, turing-forward/wavepipe); fixer report(s) not on main: fix-004, fix-007.	open
MEDIUM	**Pipeline steps failed in tracked runs** Detector: `steps.status = 'failed'`	publish ×4 (all at the signing/publish step).	open
RESOLVED	**Fabricated 'empirical' curves quarantined** Detector: `Problem-Scout candidate BIO-TPL-DQ-01, verified by fix-001`	Three bioelectric_polarity 'empirical' series were confirmed slot-seeded alpha templates and quarantined with provenance labels (PR #271); kept here as the worked example of a flag closing.	resolved

IX. The unranked fleets — a recording gap

The 1,030 slot dispatches and the Cursor cloud fleets (100 scouts, 12 fixers) do not record which model each agent ran, so they cannot be ranked yet — the leaderboard above covers only the 106 telemetry-tracked loops. Fleet outcomes by completion are in Plate I; recording the model in every dispatch record (the fixer dispatcher already accepts `--model`) would make the next wave rankable end-to-end.

X. On Deposit — plates from the natural history museum

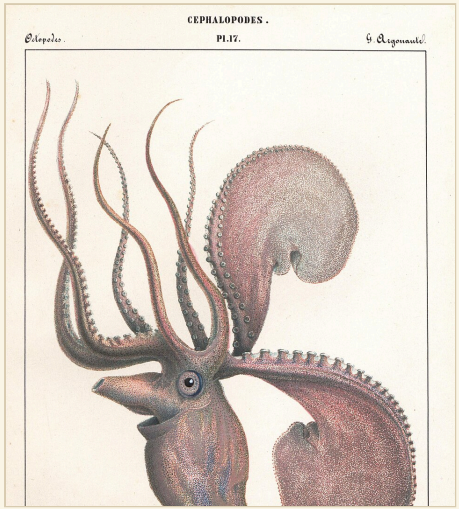


Plate XVII

Argonauta argo Linné

Vérany, Céphalopodes (1851), Pl. 17 — del. ex vivo

Exhibited with intent: an absence, carefully noted, is also a specimen.

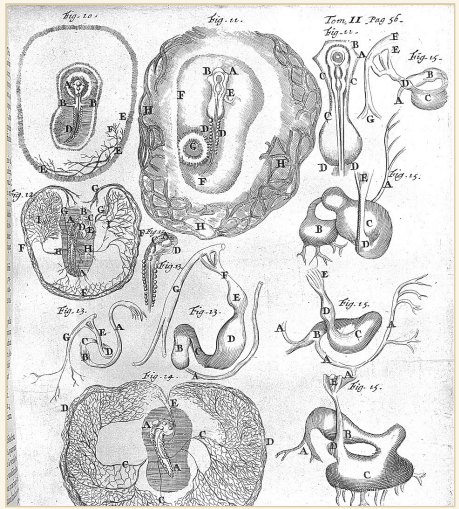
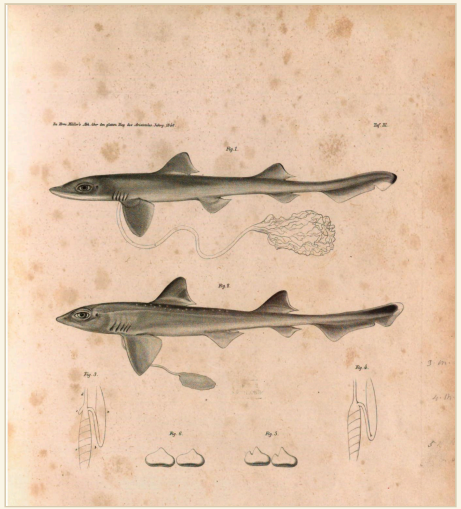


Fig. 10–15

De pulli in ovo — the chick in the egg

Malpighi (1687), Tom. II, p. 56

Development drawn figure by figure, letter by letter. Every loop still files its evidence this way.

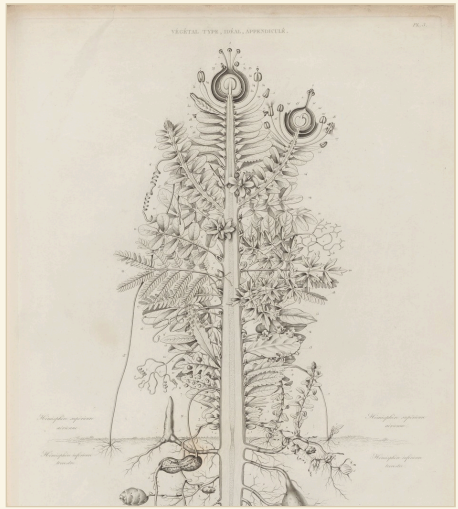


Taf. III

Mustelus laevis — the yolk-sac placenta

Johannes Müller (1842), Taf. III

Aristotle described it; for two thousand years it was thought an error, until Müller dissected it. Patron specimen for trusting careful observation over received opinion.



Pl. 3

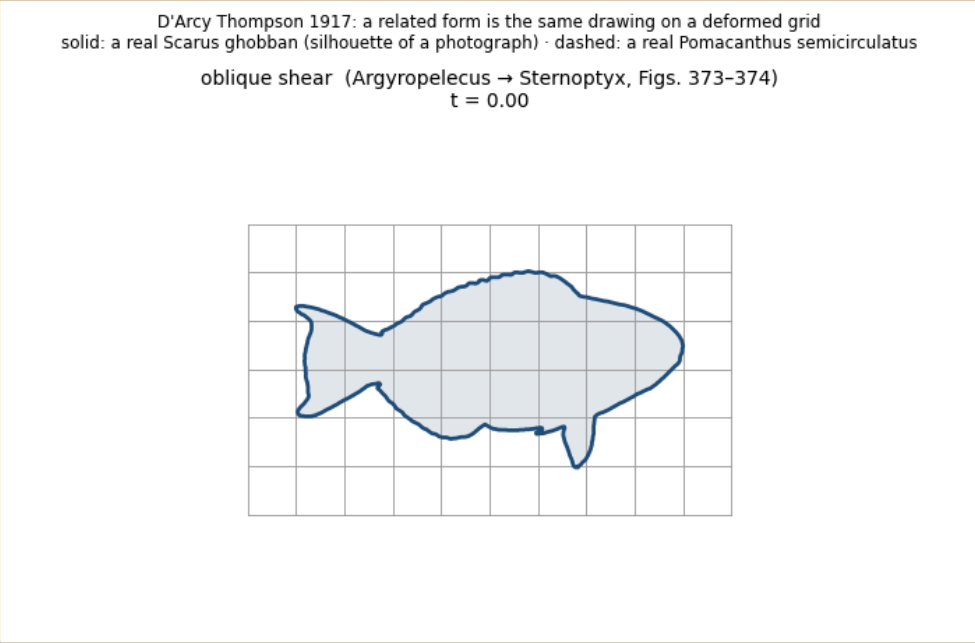
Végétal type, idéal, appendiculé — the Urpflanze

Turpin (1837), after Goethe

Hung as a standing instruction: exhibit types, reserve judgement on descent until the morphology is in hand.

Plates held in this repo at `docs/media/`, as catalogued in the museum-illustrated Thompson (1913) edition. Species portraits above are tracked library photographs from `datasets/species_gallery/` (iconic atlas), staged into `docs/media/specimens/` at build time.

XI. The Evolution Gallery — in preparation



Not yet accessioned

On Growth and Form, ch. XVII — transformations

Thompson (1917)

This room is roped off. Forms in this collection will be allowed to evolve into one another — by coordinate transformation, not by costume — when enough morphology has been captured to carry the deformation honestly. The specimens above are being pinned for precisely that day.

* Slot-agent Lean/nanopub counts are as recorded at dispatch time; per issue #268 the overnight-slot Lean modules are placeholder templates, not mechanism proofs, so treat `lean_theorems_proved`-style tallies from those records as claims, not verifications. Telemetry runs span 2026-10-18T18:03:18.181Z – 2026-10-20T02:35:07.105Z.